1. For what value of k will $(k+9), (2k-1)$ and $(2k+7)$ be consecutive terms of an A.P.?

2. The sums of the first n terms of three arithmetic progressions are S_0, S_2 and S_3 respectively. The first term of each A.P. is 1 and their common differences are $1, 2$ and 3 respectively. Prove that $S_1 + S_3 = 2S_2$.